

OpenCart

PC Configurator

From Browser to Build.

Custom OpenCart development for a Ukrainian gaming hardware retailer — a real-time PC configurator, compatibility engine, and a checkout flow built for serious buyers.

5,200+

Active SKUs

3

Custom Modules

8wk

Delivery

PC BUILD CONFIGURATOR

REAL-TIME

☒	CPU	Intel Core i9-13900K	14 500 ₺
☒	MOTHERBOARD	ASUS ROG STRIX Z790-E	9 800 ₺
☒	RAM	Corsair 32GB DDR5-5600	4 200 ₺
☐	GPU	Selecting compatible...	
☐	PSU	After GPU selection	
☐	CASE	By form factor	

Current total

28 500 ₺

Complete Build →

Customers were lost. Carts were abandoned.

Game-X had 5,200+ SKUs across CPUs, GPUs, RAM, motherboards, cases, and PSUs — but no guidance. Buyers couldn't confidently pick compatible parts. Support was overwhelmed. Conversion suffered.



No compatibility filtering

Customers ordered parts that didn't fit their build



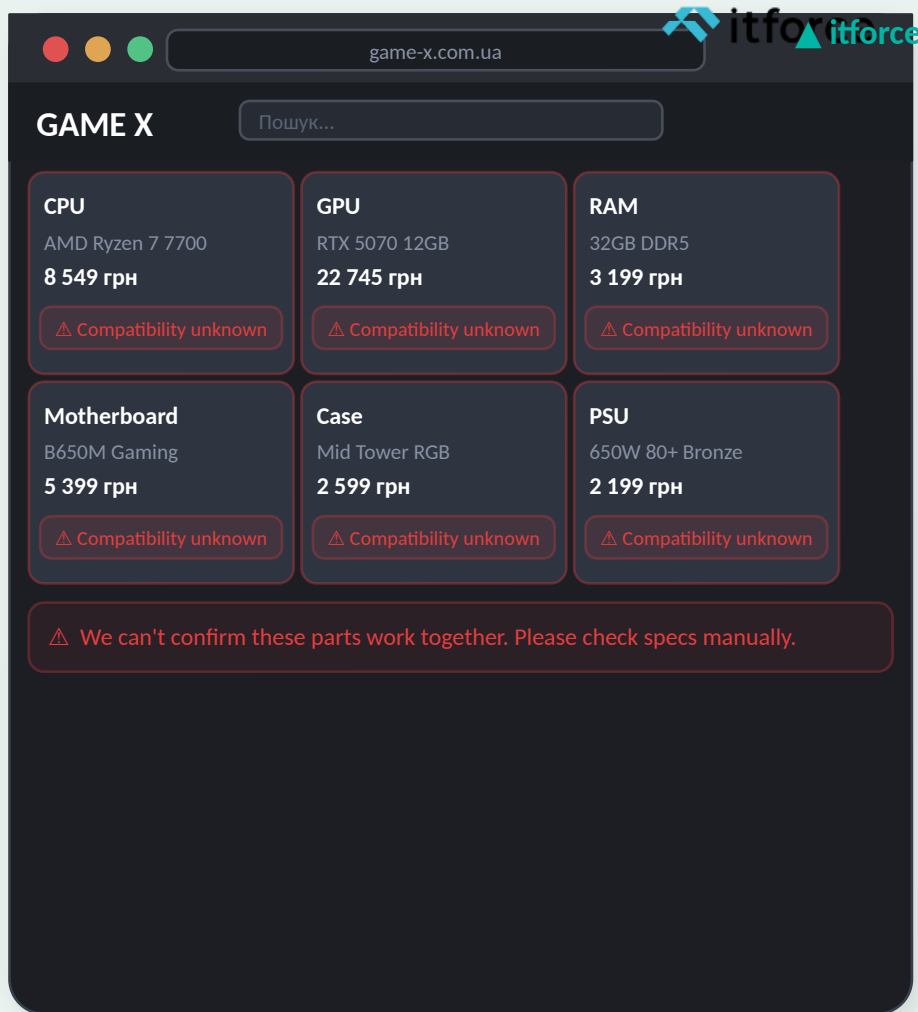
No guided flow

Product pages in isolation, no cross-sell logic



No configurator

Competitors had one. Game-X didn't.



Three custom modules. One seamless engine.

We extended OpenCart with three interconnected modules — built from scratch, not off-the-shelf.



PC Configurator

[Step-by-step guided builder](#)

6-step flow:
CPU→MB→RAM→GPU→PSU→Case.
Each step auto-filters by compatibility.
Live price counter. Mobile-responsive.



Compatibility Engine

[Real-time validation](#)

Cross-references a relational matrix: CPU socket, RAM type, GPU wattage, case form factor. Zero invalid builds reach checkout.



Smart Cart

[Build-aware checkout](#)

Saves full PC builds, detects swapped parts that break compatibility, warns buyer before checkout with plain-language explanations.

A builder that thinks like a technician.

- **CPU → MB → RAM → GPU → PSU → Case** — 6-step flow, each step locked until prior is complete
- **Auto-filters by compatibility** — Only compatible options shown — no wrong combos possible
- **Live price counter** — Updates as each component is selected
- **"Why locked?" tooltip** — Plain-language explanation for confused buyers
- **Mobile-responsive** — Full touch-optimised, thumb-friendly layout

game-x.com.ua / pc-builder

Build Your PC — Step by Step

CPU	Intel Core i9-13900K — 3.0GHz 24-core	14 500 ₺
MOTHERBOARD	ASUS ROG STRIX Z790-E — LGA1700	9 800 ₺
RAM	Corsair Vengeance 32GB DDR5-5600	4 200 ₺
GPU	Selecting compatible models...	
PSU	Auto-calculated after GPU	
CASE	Filtered by form factor	
Total		28 500 ₺

[Complete Build →](#)

5,200 SKUs.

Zero wrong combos.

The engine cross-references a relational compatibility matrix. Rules cover:

1. CPU socket ↔ Motherboard socket
2. RAM type ↔ Motherboard slot (DDR4/DDR5)
3. GPU power draw ↔ PSU wattage
4. Case form factor ↔ Motherboard size

4

Compatibility Layers

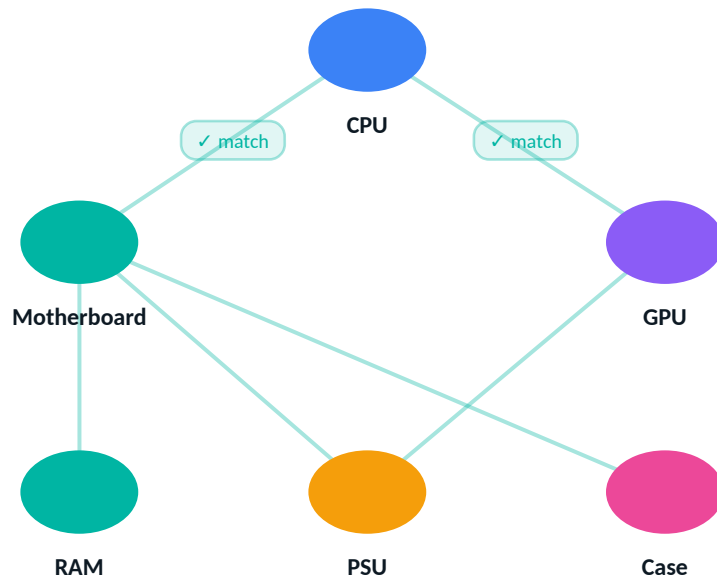
Real-time

Validation

0

Invalid Builds

Compatibility Matrix



Built to convert. Proved it fast.

6 months post-launch vs. pre-launch baseline

+34%

Conversion Rate

Configurator checkouts vs.
standard product pages

-60%

Support Tickets

Pre-sale queries after
launching the configurator

2.4×

Avg. Order Value

Configurator builds vs.
single-item purchases

4.9/5

Satisfaction Score

Post-purchase rating for
configurator orders

“

We expected a product
page upgrade.

**ITForce built us an actual
sales system.**

— Game-X founder

game-x.com.ua · Kyiv, Ukraine · OpenCart



4.9/5 customer satisfaction

✓ Completed on Upwork · Top Rated Agency · 100% Job Success

OpenCart, extended the right way.

OpenCart 3.x

PHP 8.1

MySQL

Vanilla JS

Custom OC API

CSS3

Git / CI

What we delivered

- PC Configurator — 6-step guided builder
- Compatibility Engine — relational matrix, 4 rule layers
- Smart Cart — build-save + pre-checkout validation
- Full OpenCart theme integration & admin UI

Delivery Timeline

8 weeks · discovery to production

Wk 1-2

Discovery & Schema

Compatibility matrix design, relational DB structure

Wk 3-4

Configurator Module

6-step builder + admin UI for product mapping

Wk 5-6

Compatibility Engine

Real-time validation + Smart Cart integration

Wk 7

QA & Testing

Cross-browser, mobile, edge-case compatibility

Wk 8

Deploy & Handover

Staging → production + Game-X team training